TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Capstone Design Project I | 2026–2027

PerDrobe Project Proposal

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Project name PerDrobe

Project webpage https://perdrobe2026-commits.github.io/

Team Members

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Project Description

PerDrobe is a personalized wardrobe and clothing recommendation app we plan to develop for our CMPE 491–492 project. We want to make it easier for people to put together outfits using clothes they already own, especially when they find a style they like but are unsure how to recreate it. Users will be able to add their clothes to a private digital wardrobe and choose their preferred styles and fits. Their likes, saved outfits, and followed accounts will also help us understand their preferences. With Outfit Remix, users can replace pieces from an outfit they discover with suitable items from their own wardrobe. The app will also recommend new clothes that match what they already have. We plan to provide size suggestions using optional body measurements, usual sizes, and available size charts, but only when there is enough information to make a reasonable recommendation.

The main technical part of our project is an AI-based recommendation system that finds suitable clothing items, checks how well they work together, and ranks them based on each user's preferences. We plan to compare three approaches. The first will use clothing attributes, the second will also consider user preferences and interactions, and the third will include the user's existing wardrobe. We will evaluate how relevant and useful the recommendations are, how much variety they offer, how well they cover the available items, and how quickly the system responds. We will also check size and fit recommendations when suitable reference data is available. Since the project is still in the planning stage, we do not have any results to report yet. If we complete the main features on time, we may also add Duo Match and image-based features.